

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT
on

Object Oriented Java Programming **(23CS3PCOOJ)**

Submitted by

Dhruvdeep Nayak (1BM23CS093)

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)

BENGALURU-560019
Sep-2024 to Jan-2025

B.M.S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Object Oriented Java Programming (23CS3PCOOJ)” carried out by **Dhruvdeep Nayak (1BM23CS093)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Object-Oriented Java Programming (23CS3PCOOJ) work prescribed for the said degree.

Dr Surabhi S Assistant Professor Department of CSE, BMSCE	Dr Kavita Sooda Professor & HOD Department of CSE, BMSCE
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I N D E X				
Name <u>DHRUVDEEP NAVAK</u>		Std _____	Sec _____	
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1	1st Oct	Add and subtract two numbers		
2	1st Oct	Product of two numbers		
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12	3rd Dec	Threads	(10)	28.03.12
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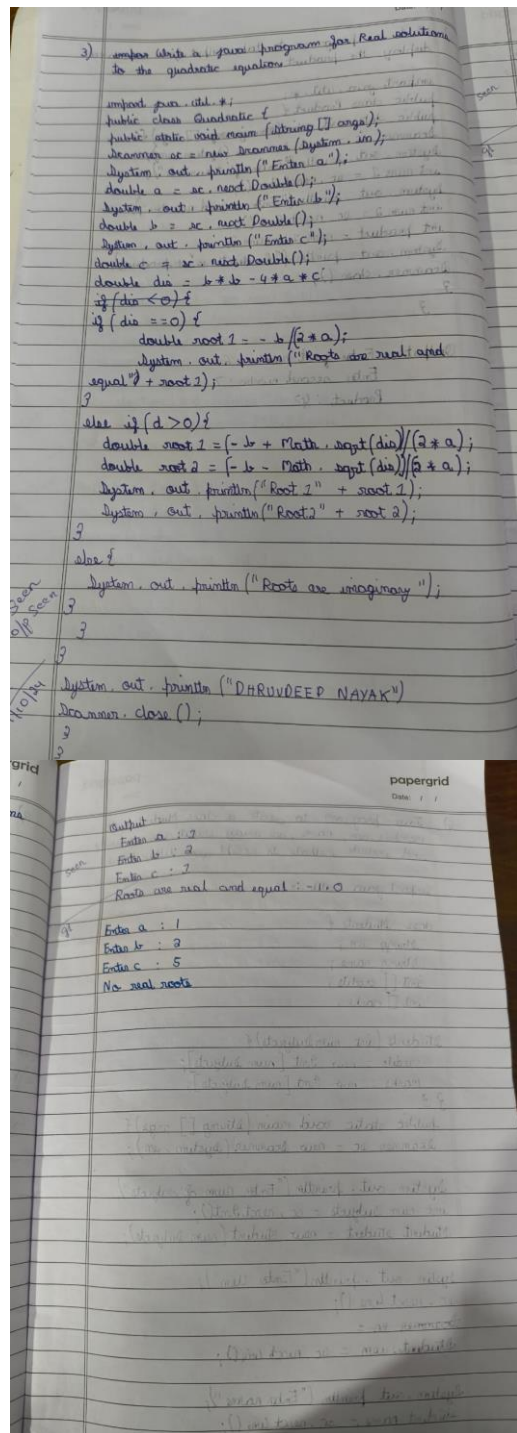
Github Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program1>

Program 1:

Develop a Java program that prints all real solutions to the quadratic equation $ax^2 + bx + c = 0$. Read in a, b, c and use the quadratic formula. If the discriminate $b^2 - 4ac$ is negative, display a message stating that there are no real solutions.

Observation Book:



Code:

import java.util.Scanner;

public class Quadratic {

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.println("Enter a:");
    double a = sc.nextDouble();

    System.out.println("Enter b:");
    double b = sc.nextDouble();

    System.out.println("Enter c:");
    double c = sc.nextDouble();

    double dis = b * b - 4 * a * c;

    if (dis < 0) {
        System.out.println("Roots are imaginary");
    } else if (dis == 0) {
        double root1 = -b / (2 * a);
        System.out.println("Roots are real and equal: " + root1);
    } else {
        double root1 = (-b + Math.sqrt(dis)) / (2 * a);
        double root2 = (-b - Math.sqrt(dis)) / (2 * a);
        System.out.println("Roots are real and distinct:");
        System.out.println("Root 1: " + root1);
        System.out.println("Root 2: " + root2);
    }

    System.out.println("DHRUVDEEP NAYAK");
    sc.close();
}
}

```

Output

```
C:\windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac Main.java

C:\Users\dhruv\OneDrive\Documents\Java>java Main
Enter a:
1
Enter b:
2
Enter c:
1
Roots are real and equal: -1.0
DHRUVDEEP NAYAK

C:\Users\dhruv\OneDrive\Documents\Java>java Main
Enter a:
1
Enter b:
2
Enter c:
5
Roots are imaginary
DHRUVDEEP NAYAK
```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program2>

Program 2:

Develop a Java program to create a class Student with members usn, name, an array credits and an array marks. Include methods to accept and display details and a method to calculate SGPA of a student.

Observation Book:

```

4) Java program to create a class Student with
   marks, roll no, name, an array credits, array marks
   and include methods to accept and display details.

import java.util.*;

class Student {
    String roll;
    String name;
    int[] credits;
    int[] marks;

    Student(int numSubjects) {
        credits = new int[numSubjects];
        marks = new int[numSubjects];
    }

    // public static void main (String[] args) {
    //     Scanner sc = new Scanner(System.in);

    //     System.out.println("Enter num of subjects");
    //     int numSubjects = sc.nextInt();

    //     Student student = new Student(numSubjects);

    //     System.out.println("Enter roll no");
    //     sc.nextLine();
    //     Scanner sc = new Scanner(System.in);
    //     student.roll = sc.nextLine();

    //     System.out.println("Enter name");
    //     student.name = sc.nextLine();

    //     for(int i = 0; i < numSubjects; i++) {

```

```

        System.out.println("Enter credits for subject " + (i+1));
        student.credits[i] = sc.nextInt();
        System.out.println("Enter marks");
        student.marks[i] = sc.nextInt();
    }

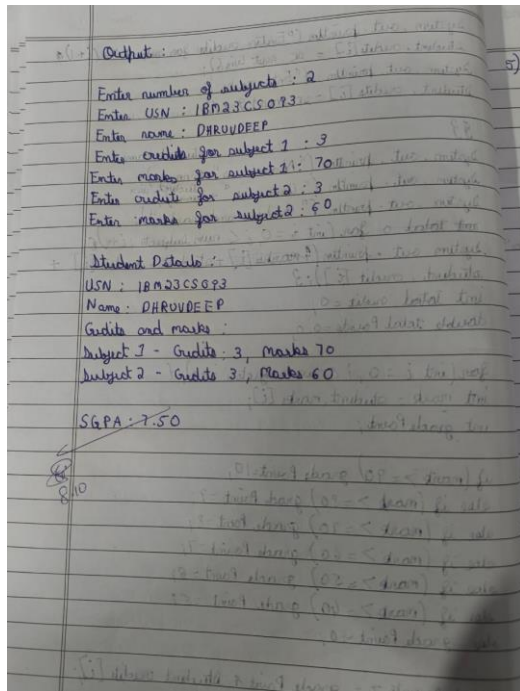
    // public void displayStudentDetails() {
    //     System.out.println("Name is " + name);
    //     System.out.println("Roll no is " + roll);
    //     System.out.println("Credits and marks");
    //     int total = 0;
    //     for(int i = 0; i < marks.length; i++) {
    //         System.out.println("marks[" + i + "] + student.credits[i] +
    //         student.credits[i];
    //     }
    //     int totalCredits = 0;
    //     double totalPoints = 0.0;
    //     for(int i = 0; i < marks.length; i++) {
    //         int mark = student.marks[i];
    //         int gradePoint;

    //         if (mark >= 90) gradePoint = 10;
    //         else if (mark >= 80) gradePoint = 9;
    //         else if (mark >= 70) gradePoint = 8;
    //         else if (mark >= 60) gradePoint = 7;
    //         else if (mark >= 50) gradePoint = 6;
    //         else if (mark >= 40) gradePoint = 5;
    //         else gradePoint = 0;

    //         totalPoints += gradePoint * student.credits[i];
    //         totalCredits += student.credits[i];
    //     }
    //     double sgpa = totalPoints / totalCredits;
    //     System.out.println("Sgpa is: " + sgpa);
    // }
}

// Compiler
// 8.10.24

```

Code:

```
import java.util.Scanner;

class Student {
    String usn;
    String name;
    int[] credits;
    int[] marks;

    Student(int numSubjects) {
        credits = new int[numSubjects];
        marks = new int[numSubjects];
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number of subjects:");
        int numSubjects = sc.nextInt();

        Student student = new Student(numSubjects);
```

```

System.out.println("Enter USN:");
student.usn = sc.next();

System.out.println("Enter name:");
sc.nextLine();
student.name = sc.nextLine();

for (int i = 0; i < numSubjects; i++) {
    System.out.println("Enter credits for subject " + (i + 1) + ":");
    student.credits[i] = sc.nextInt();

    System.out.println("Enter marks for subject " + (i + 1) + ":");
    student.marks[i] = sc.nextInt();
}

System.out.println("\nStudent Details:");
System.out.println("Name: " + student.name);
System.out.println("USN: " + student.usn);
System.out.println("Credits and Marks:");
for (int i = 0; i < numSubjects; i++) {
    System.out.println("Subject " + (i + 1) + " - Credits: " + student.credits[i] + ", Marks: " +
student.marks[i]);
}

int totalCredits = 0;
double totalPoints = 0.0;

for (int i = 0; i < numSubjects; i++) {
    int mark = student.marks[i];
    int gradePoint;

    if (mark >= 90) gradePoint = 10;
    else if (mark >= 80) gradePoint = 9;
    else if (mark >= 70) gradePoint = 8;
    else if (mark >= 60) gradePoint = 7;
    else if (mark >= 50) gradePoint = 6;
    else if (mark >= 40) gradePoint = 5;
    else gradePoint = 0;

    totalPoints += gradePoint * student.credits[i];
    totalCredits += student.credits[i];
}

double sgpa = totalPoints / totalCredits;

System.out.printf("SGPA is: %.2f\n", sgpa);

```

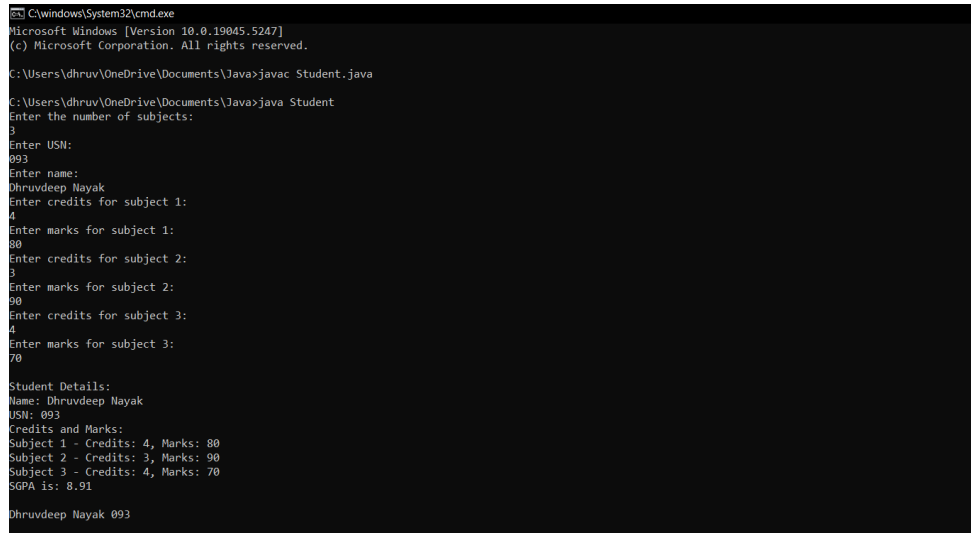
```

        System.out.println("\nDhruvdeep Nayak 093");

        sc.close();
    }
}

```

Output:



```

C:\windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac Student.java

C:\Users\dhruv\OneDrive\Documents\Java>java Student
Enter the number of subjects:
3
Enter USN:
093
Enter name:
Dhruvdeep Nayak
Enter credits for subject 1:
4
Enter marks for subject 1:
80
Enter credits for subject 2:
3
Enter marks for subject 2:
90
Enter credits for subject 3:
4
Enter marks for subject 3:
70

Student Details:
Name: Dhruvdeep Nayak
USN: 093
Credits and Marks:
Subject 1 - Credits: 4, Marks: 80
Subject 2 - Credits: 3, Marks: 90
Subject 3 - Credits: 4, Marks: 70
SGPA is: 8.91

Dhruvdeep Nayak 093

```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program3>

Program 3:

Create a class Book which contains four members: name, author, price, num_pages. Include a constructor to set the values for the members. Include methods to set and get the details of the objects. Include a toString() method that could display the complete details of the book. Develop a Java program to create n book objects.

Observation Book

Q7) Create a class Book which contains four attributes: name, author, price, num. pages. Include a constructor to set the values for members. Include methods to set and get details of the objects. Include methods to set and get details.

```

import java.util.*;

class Book {
    String name;
    String author;
    double price;
    int numPages;

    public Book(String name, String author, double price, int numPages) {
        this.name = name;
        this.author = author;
        this.price = price;
        this.numPages = numPages;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getName() {
        return name;
    }

    public void setAuthor(String author) {
        this.author = author;
    }

    public String getAuthor() {
        return author;
    }
}

```

```

    public void setPrice(double price) {
        this.price = price;
    }

    public double getPrice() {
        return price;
    }

    public void setNumPages(int numPages) {
        this.numPages = numPages;
    }

    public String toString() {
        return "Book Name: " + name + " Author: " + author +
            " Price: " + price + " Number of Pages: " + numPages;
    }

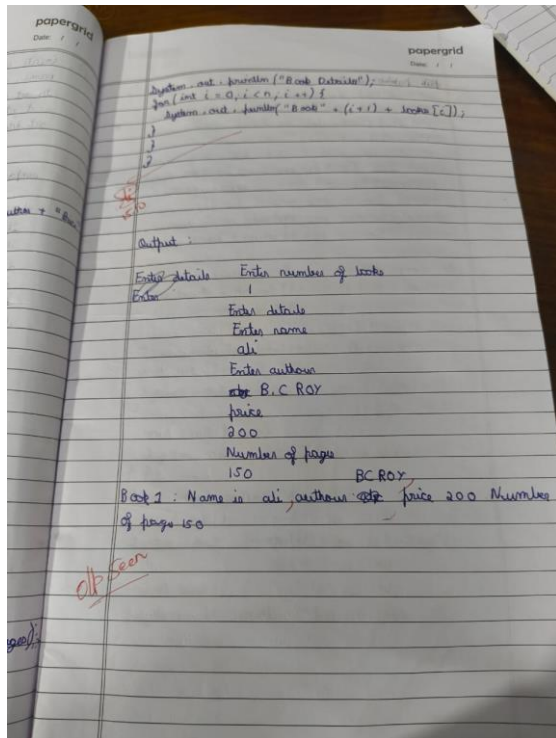
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter number of books:");
        int n = sc.nextInt();
        sc.nextLine();

        Book[] books = new Book[n];

        for (int i = 0; i < n; i++) {
            System.out.println("Enter details");
            String name = sc.nextLine();
            String author = sc.nextLine();
            System.out.println("Price");
            double price = sc.nextDouble();
            System.out.println("Number of Pages");
            int numPages = sc.nextInt();

            books[i] = new Book(name, author, price, numPages);
        }
    }
}

```



Code:

```
import java.util.Scanner;
```

```
class Book {
```

```
    String name;
```

```
    String author;
```

```
    double price;
```

```
    int numPages;
```

```
    public Book(String name, String author, double price, int numPages) {
```

```
        this.name = name;
```

```
        this.author = author;
```

```
        this.price = price;
```

```
        this.numPages = numPages;
```

```
    }
```

```
    public void setName(String name) {
```

```
        this.name = name;
```

```
    }
```

```
    public String getName() {
```

```

        return name;
    }

    public void setAuthor(String author) {
        this.author = author;
    }

    public String getAuthor() {
        return author;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public double getPrice() {
        return price;
    }

    public void setNumPages(int numPages) {
        this.numPages = numPages;
    }

    public int getNumPages() {
        return numPages;
    }

    public String toString() {
        return "Book Name: " + name + ", Author: " + author + ", Price: " + price + ", Number of Pages: " + numPages;
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter number of books:");
        int n = sc.nextInt();
        sc.nextLine();

        Book[] books = new Book[n];

        for (int i = 0; i < n; i++) {
            System.out.println("Enter details for Book " + (i + 1) + ":");
            System.out.println("Enter name:");
            String name = sc.nextLine();

            System.out.println("Enter author:");

```

```

        String author = sc.nextLine();

        System.out.println("Enter price:");
        double price = sc.nextDouble();

        System.out.println("Enter number of pages:");
        int numPages = sc.nextInt();
        sc.nextLine();

        books[i] = new Book(name, author, price, numPages);
    }

    System.out.println("Book Details:");
    for (int i = 0; i < n; i++) {
        System.out.println("Book " + (i + 1) + ": " + books[i]);
    }

    System.out.println("Dhruvdeep Nayak-093");
    sc.close();
}
}

```

```

C:\windows\System32\cmd.exe
C:\Users\dhruv\OneDrive\Documents\Java>javac Book.java
C:\Users\dhruv\OneDrive\Documents\Java>java Book
Enter number of books:
2
Enter details for Book 1:
Enter name:
basic of java
Enter author:
RV Ram
Enter price:
400
Enter number of pages:
250
Enter details for Book 2:
Enter name:
basic of python
Enter author:
BC Krishna
Enter price:
500
Enter number of pages:
600
Book Details:
Book 1: Book Name: basic of java, Author: RV Ram, Price: 400.0, Number of Pages: 250
Book 2: Book Name: basic of python, Author: BC Krishna, Price: 500.0, Number of Pages: 600
Dhruvdeep Nayak-093
C:\Users\dhruv\OneDrive\Documents\Java>

```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program%204>

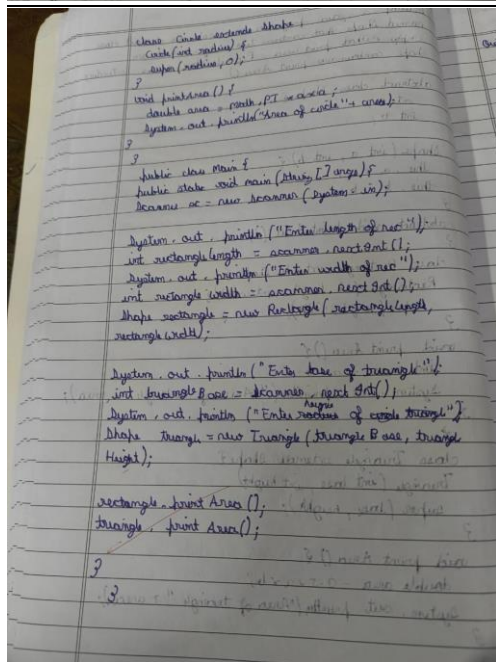
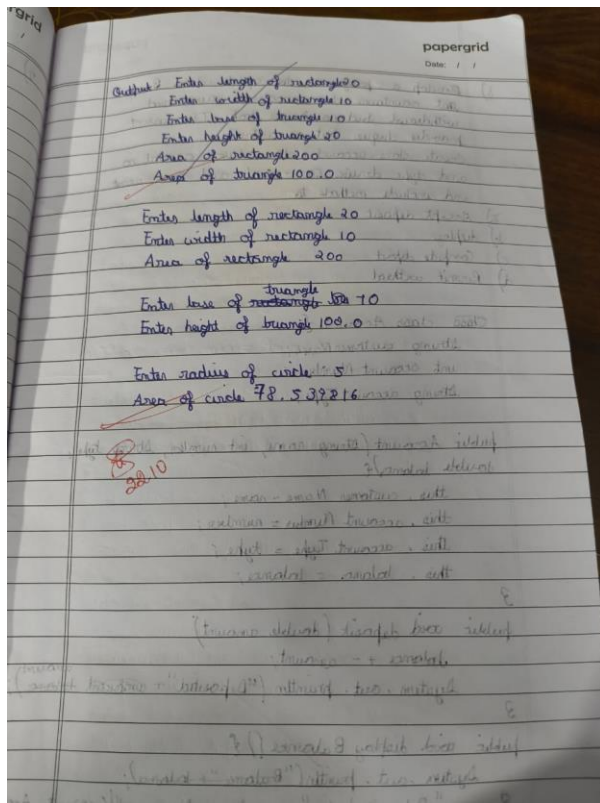
Program 4:

Develop a Java program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea() that prints the area of the given shape.

Observation Book:

7) Develop a Java program to create an abstract class named Shape that contains two integers and an empty method printArea(). Provide three classes Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea() that prints the area of the given shape.

```
abstract class Shape {  
    int a;  
    int b;  
  
    Shape(int a, int b) {  
        this.a = a;  
        this.b = b;  
    }  
  
    abstract void printArea();  
  
    class Rectangle extends Shape {  
        Rectangle(int length, int width) {  
            super(length, width);  
        }  
  
        void printArea() {  
            int area = a * b;  
            System.out.println("Area of Rectangle" + area);  
        }  
  
        class Triangle extends Shape {  
            Triangle(int base, int height) {  
                super(base, height);  
            }  
  
            void printArea() {  
                double area = 0.5 * a * b;  
                System.out.println("Area of Triangle" + area);  
            }  
        }  
    }  
}
```

CODE:

```
import java.util.Scanner;
```

```
abstract class Shape {
```

```
    int a;
```

```
    int b;
```

```

    Shape(int a, int b) {
        this.a = a;
        this.b = b;
    }

    abstract void printArea();
}

class Rectangle extends Shape {
    Rectangle(int length, int width) {
        super(length, width);
    }

    void printArea() {
        int area = a * b;
        System.out.println("Area of Rectangle: " + area);
    }
}

class Triangle extends Shape {
    Triangle(int base, int height) {
        super(base, height);
    }

    void printArea() {
        double area = 0.5 * a * b;
        System.out.println("Area of Triangle: " + area);
    }
}

class Circle extends Shape {
    Circle(int radius) {
        super(radius, 0);
    }

    void printArea() {
        double area = Math.PI * a * a;
        System.out.println("Area of Circle: " + area);
    }
}

public class Shapecalc {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter length of rectangle:");
    }
}

```

```

int rectangleLength = scanner.nextInt();

System.out.println("Enter width of rectangle:");
int rectangleWidth = scanner.nextInt();
Shape rectangle = new Rectangle(rectangleLength, rectangleWidth);

System.out.println("Enter base of triangle:");
int triangleBase = scanner.nextInt();

System.out.println("Enter height of triangle:");
int triangleHeight = scanner.nextInt();
Shape triangle = new Triangle(triangleBase, triangleHeight);

System.out.println("Enter radius of circle:");
int circleRadius = scanner.nextInt();
Shape circle = new Circle(circleRadius);

rectangle.printArea();
triangle.printArea();
circle.printArea();

System.out.println("Dhruvdeep Nayak-093");

scanner.close();
}
}

```

Output:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac Shapecalc.java

C:\Users\dhruv\OneDrive\Documents\Java>java Shapecalc
Enter length of rectangle:
20
Enter width of rectangle:
10
Enter base of triangle:
2
Enter height of triangle:
4
Enter radius of circle:
3
Area of Rectangle: 200
Area of Triangle: 4.0
Area of Circle: 28.274333882308138
Dhruvdeep Nayak-093

C:\Users\dhruv\OneDrive\Documents\Java>
```

Github Link

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program5>

Program 5:

Develop a Java program to create a class Bank that maintains two kinds of account for its customers, one called savings account and the other current account. The savings account provides compound interest and withdrawal facilities but no cheque book facility. The current account provides cheque book facility but no interest. Current account holders should also maintain a minimum balance and if the balance falls below this level, a service charge is imposed. Create a class Account that stores customer name, account number and type of account. From this derive the classes Cur-acct and Sav-acct to make them more specific to their requirements. Include the necessary methods in order to achieve the following tasks: a) Accept deposit from customer and update the balance. b) Display the balance. c) Compute and deposit interest d) Permit withdrawal and update the balance Check for the minimum balance, impose penalty if necessary and update the balance.

Observation Book:

8) Develop a program to create class Bank that maintains two accounts saving and withdraw but no cheque book. The current provides cheque but no interest. Create class account store name, account no and type derive cur-accnt and sav-accnt and include methods to

- Accept deposit
- display
- Compute interest
- Withdraw

```

class class AccountName;
String customerName;
int account Number;
String account Type;

public Account(String name, int number, String type, double balance) {
    this.customerName = name;
    this.accountNumber = number;
    this.accountType = type;
    this.balance = balance;
}

public void deposit(double amount) {
    balance += amount;
    System.out.println("Deposited " + amount + " amount balance");
}

public void displayBalance() {
    System.out.println("Balance " + balance);
    "Deposited by " + customerName + " Current Account balance " + balance
    
```

```

import java.util.*;
class Sav - accnt extends Account {
    double minimumBalance;
    double interestCharge;

    public Sav - accnt(String name, int number, double balance, double minBalance, double interestCharge) {
        super(name, number, "Savings", balance);
        this.minimumBalance = minBalance;
        this.interestCharge = interestCharge;
    }

    public void withdraw(double amount) {
        if (balance - amount < minimumBalance) {
            balance -= interestCharge;
            System.out.println("Balance Change " + interestCharge);
        } else {
            balance -= amount;
            System.out.println("Amount withdrawn " + balance);
        }
    }
}

class Sav - accnt extends Account {
    double interestRate;

    public Sav - accnt(String name, int number, double balance, double interestRate) {
        super(name, number, "Savings", balance);
        this.interestRate = interestRate;
    }

    public void depositInterest() {
        double interest = balance * interestRate / 100;
        balance += interest;
    }
}
    
```

```

system.out.println("Interest added" + interest);

public class Bank {
    public static void main(String[] args) {
        int acct;
        currentAccount = new CurrentAccount(1000, 1000, 50);
        currentAccount.deposit(1000);
        currentAccount.displayBalance();
        currentAccount.withdraw(4500);
        currentAccount.displayBalance();

        new SavingsAccount(1000, 1000, 50);
        SavingsAccount.deposit(500);
        SavingsAccount.depositInterest();
        SavingsAccount.displayBalance();
    }
}

```

Output:

```

Deposited 1000
Balance 6000
Withdrawn 4500
Balance 1500
Interest added 125.0
Balance 2625.0

```

John Deposited 1000.0 in Current Account
 Deposited by John / Current Account Balance 6000
 Balance 6000.0
 John Withdrawn 4500
 Balance : 1500
 John Balance 1500.0
 Jane Deposited 500.0 in Savings
 Jane Balance 2500.0 in Savings
 Interest added 125.0
 Jane Balance 2625.0 in Savings

John : Current Balance 6000.0 Current Account
 Jane : Current Balance 2625.0 Savings Account

29.10 Display the output

Code:

```

import java.util.*;

class cur_acct extends account {
    double minimumBalance;
    double serviceCharge;

    public cur_acct(String name, int number, double balance, double minBalance, double
serviceCharge) {
        super(name, number, "Current", balance);
        this.minimumBalance = minBalance;
        this.serviceCharge = serviceCharge;
    }

    public void withdraw(double amount) {
        if (balance - amount < minimumBalance) {
            balance -= serviceCharge;
            System.out.println("Service Charge: " + serviceCharge);
        }
        balance -= amount;
        System.out.println("Amount withdrawn by " + customerName + ": " + amount + ", Remaining
Balance: " + balance);
    }
}

class sav_acct extends account {
    double interestRate;

    public sav_acct(String name, int number, double balance, double interestRate) {
        super(name, number, "Savings", balance);
        this.interestRate = interestRate;
    }

    public void depositInterest() {
        double interest = balance * interestRate / 100;
        balance += interest;
        System.out.println("Interest added to " + customerName + "'s account: " + interest);
    }
}

class account {
    String customerName;
    int accountNumber;
    String accountType;
    double balance;

    public account(String name, int number, String type, double balance) {

```

```

        this.customerName = name;
        this.accountNumber = number;
        this.accountType = type;
        this.balance = balance;
    }

    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited by " + customerName + ": " + amount);
    }

    public void displayBalance() {
        System.out.println("Balance for " + customerName + ": " + balance);
    }
}

public class Bank {
    public static void main(String[] args) {
        cur_acct currentAccount = new cur_acct("James", 12345, 5000, 1000, 50);
        currentAccount.deposit(1000);
        currentAccount.displayBalance();
        currentAccount.withdraw(4500);
        currentAccount.displayBalance();

        sav_acct savingsAccount = new sav_acct("Judy", 67890, 2000, 5);
        savingsAccount.deposit(500);
        savingsAccount.depositInterest();
        savingsAccount.displayBalance();

        System.out.println("Dhruvdeep Nayak-093");
    }
}

```

Output:


```
C:\windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac Bank.java

C:\Users\dhruv\OneDrive\Documents\Java>java Bank
Deposited by James: 1000.0
Balance for James: 6000.0
Amount withdrawn by James: 4500.0, Remaining Balance: 1500.0
Balance for James: 1500.0
Deposited by Judy: 500.0
Interest added to Judy's account: 125.0
Balance for Judy: 2625.0
Dhruvdeep Nayak-093

C:\Users\dhruv\OneDrive\Documents\Java>
```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program6>

Program 6:

Create a package CIE which has two classes- Student and Internals. The class Personal has members like usn, name, sem. The class internals has an array that stores the internal marks scored in five courses of the current semester of the student. Create another package SEE which has the class External which is a derived class of Student. This class has an array that stores the SEE marks scored in five courses of the current semester of the student. Import the two packages in a file that declares the final marks of n students in all five courses.

Observation Book:

Create a package CIE which has two classes Student and External. class Personal with personal details class Internal has an array that stores internal marks. Create package SEE with class External and implement the two packages and display final marks

```
package CIE;
public class Personal {String un; name; int age;}
```

```
package CIE;
public class Internal extends Personal {int [] internalMarks;
= new int[5];
int [] internalMarks = new int[5];
}
```

```
package SEE;
import CIE, Internal;
public class External extends Internal {
int [] externalMarks = new int[5];
}
```

```
import CIE.*;
import SEE.*;
```

```
public class Final Marks {
public static void main(String [] args){
Scanner sc = new Scanner(System.in);
System.out.println("Enter no of students");
int n = sc.nextInt();
External [] students = new External[n];
for (int i = 0; i < n; i++)
```


Code:

```
package CIE;

public class Personal {
    String name;
    String usn;
    int sem;

    public Personal(String name, String usn, int sem) {
        this.name = name;
        this.usn = usn;
        this.sem = sem;
    }

    public String getName() {
        return name;
    }

    public String getUsn() {
        return usn;
    }

    public int getSem() {
        return sem;
    }
}

public class Internals {
    Personal personalDetails; // Contains personal details
    int[] internalMarks = new int[5]; // Stores 5 internal marks

    public Internals(Personal personalDetails) {
        this.personalDetails = personalDetails;
    }

    public void setInternalMarks(int[] marks) {
        for (int i = 0; i < 5; i++) {
            internalMarks[i] = marks[i];
        }
    }

    public int[] getInternalMarks() {
        return internalMarks;
    }

    public String getName() {
```

```

        return personalDetails.getName();
    }

    public String getUsn() {
        return personalDetails.getUsn();
    }

    public int getSem() {
        return personalDetails.getSem();
    }
}

package SEE;

import CIE.Internals;
import CIE.Personal;
import java.util.Scanner;

public class External extends Internals {
    int[] externalMarks = new int[5]; // Stores 5 external marks

    public External(Personal personalDetails) {
        super(personalDetails);
    }

    public void setExternalMarks(int[] marks) {
        for (int i = 0; i < 5; i++) {
            externalMarks[i] = marks[i];
        }
    }

    public int[] getExternalMarks() {
        return externalMarks;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of students: ");
        int n = sc.nextInt();

        External[] students = new External[n];

        for (int i = 0; i < n; i++) {
            System.out.println("Enter details for student " + (i + 1) + ":");

```

```

        System.out.print("Name: ");
        sc.nextLine(); // consume newline
        String name = sc.nextLine();

        System.out.print("USN: ");
        String usn = sc.nextLine();

        System.out.print("Semester: ");
        int sem = sc.nextInt();

        Personal personalDetails = new Personal(name, usn, sem);
        students[i] = new External(personalDetails);

        System.out.println("Enter 5 internal marks: ");
        int[] internalMarks = new int[5];
        for (int j = 0; j < 5; j++) {
            internalMarks[j] = sc.nextInt();
        }
        students[i].setInternalMarks(internalMarks);

        System.out.println("Enter 5 external marks: ");
        int[] externalMarks = new int[5];
        for (int j = 0; j < 5; j++) {
            externalMarks[j] = sc.nextInt();
        }
        students[i].setExternalMarks(externalMarks);
    }

    System.out.println("\nFinal Marks Details:");
    for (int i = 0; i < n; i++) {
        System.out.println("Student " + (i + 1) + ": USN = " + students[i].getUsn() + ", Name = " +
students[i].getName() + ", Semester = " + students[i].getSem());
        for (int j = 0; j < 5; j++) {
            int finalMarks = students[i].getInternalMarks()[j] + students[i].getExternalMarks()[j];
            System.out.println("Subject " + (j + 1) + " Final Marks: " + finalMarks);
        }
    }

    System.out.println("Dhruvdeep Nayak-093");

    sc.close();
}
}

```

Output:

```
C:\Users\Dell\Desktop\Check>javac Try.java

C:\Users\Dell\Desktop\Check>java Try
Enter the number of students: 2
Enter details for student 1:
Name: John
USN: 78
Semester: 3
Enter 5 internal marks:
35 33 32 28 26
Enter 5 external marks:
88 85 97 96 87
Enter details for student 2:
Name: DD
USN: 80
Semester: 4
Enter 5 internal marks:
40 39 38 40 40
Enter 5 external marks:
99 97 95 99 98

Final Marks Details:
Student 1: USN = 78, Name = John, Semester = 3
Subject 1 Final Marks: 123
Subject 2 Final Marks: 118
Subject 3 Final Marks: 129
Subject 4 Final Marks: 124
Subject 5 Final Marks: 113
Student 2: USN = 80, Name = DD, Semester = 4
Subject 1 Final Marks: 139
Subject 2 Final Marks: 136
Subject 3 Final Marks: 133
Subject 4 Final Marks: 139
Subject 5 Final Marks: 138
Dhruvdeep Nayak-093
```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program7>

Program 7:

Find Area of 3 Different shapes using Interfaces

Observation Book:

papergrid
Date: / /

Such an interface named Polygon includes default
methods: getPerimeter() and abstract class getArea().
Calculate perimeter and area of polygon.

interface Polygon {
 double perimeter; // default double
 double getArea();
}

class Rectangle implements Polygon {
 private double length; // default double
 private double width; // default double
 public Rectangle(double length, double width) {
 this.length = length;
 this.width = width;
 }
 public double getArea() {
 return length * width;
 }
}

class Triangle implements Polygon {
 private double base; // default double
 private double height; // default double
 public Triangle(double base, double height) {
 this.base = base;
 this.height = height;
 }
 public double getArea() {
 return 0.5 * base * height;
 }
}

public Triangle (double base, double height) {
 this.base = base;
 this.height = height;
}

public double getArea() {
 return 0.5 * base * height;
}

public class Rectangle {
 private double length; // default double
 private double width; // default double
 public Rectangle (double length, double width) {
 this.length = length;
 this.width = width;
 }
 public double getArea() {
 return length * width;
 }
}

System.out.println ("Enter length and width of rect");
double rectLength = scanner.nextDouble();
double rectWidth = scanner.nextDouble();
Rectangle rect = new Rectangle (rectLength, rectWidth);
System.out.println ("Enter sides of rect");
double[] rectSides = new double [4];
for (int i = 0; i < 4; i++)
 rectSides[i] = scanner.nextDouble();
System.out.println ("Rect Perimeter" + rect.getPerimeter());
System.out.println ("Rect Area" + rect.getArea());

Output:
Enter length of rect and width of rect
10 20
Enter sides of rect
10 10 20 20
Area = 200.0 Perimeter = 60.0

Code:

```
import java.util.Scanner;

interface Polygon {
    default double getPerimeter(double[] sides) {
        double perimeter = 0;
        for (double side : sides) {
            perimeter += side;
        }
        return perimeter;
    }

    double getArea();
}

class Rectangle implements Polygon {
    private double length;
    private double width;

    public Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }

    @Override
    public double getArea() {
        return length * width;
    }
}

class Triangle implements Polygon {
    private double base;
    private double height;

    public Triangle(double base, double height) {
        this.base = base;
        this.height = height;
    }

    @Override
    public double getArea() {
        return 0.5 * base * height;
    }
}

class Circle implements Polygon {
    private double radius;
```

```

public Circle(double radius) {
    this.radius = radius;
}

@Override
public double getArea() {
    return Math.PI * radius * radius;
}

// Override getPerimeter to calculate circumference for the circle
public double getPerimeter() {
    return 2 * Math.PI * radius;
}
}

public class Interface {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter length and width of rectangle:");
        double rectLength = scanner.nextDouble();
        double rectWidth = scanner.nextDouble();
        Rectangle rect = new Rectangle(rectLength, rectWidth);

        double[] rectSides = {rectLength, rectWidth, rectLength, rectWidth};
        double rectPerimeter = rect.getPerimeter(rectSides);

        System.out.println("Rectangle Perimeter: " + rectPerimeter);
        System.out.println("Rectangle Area: " + rect.getArea());

        System.out.println("\nEnter base and height of triangle:");
        double triBase = scanner.nextDouble();
        double triHeight = scanner.nextDouble();
        Triangle tri = new Triangle(triBase, triHeight);

        System.out.println("Enter the 3 sides of the triangle:");
        double[] triSides = new double[3];
        for (int i = 0; i < 3; i++) {
            triSides[i] = scanner.nextDouble();
        }
        double triPerimeter = tri.getPerimeter(triSides);

        System.out.println("Triangle Perimeter: " + triPerimeter);
        System.out.println("Triangle Area: " + tri.getArea());
    }
}

```

```

        System.out.println("\nEnter radius of circle:");
        double radius = scanner.nextDouble();
        Circle circle = new Circle(radius);

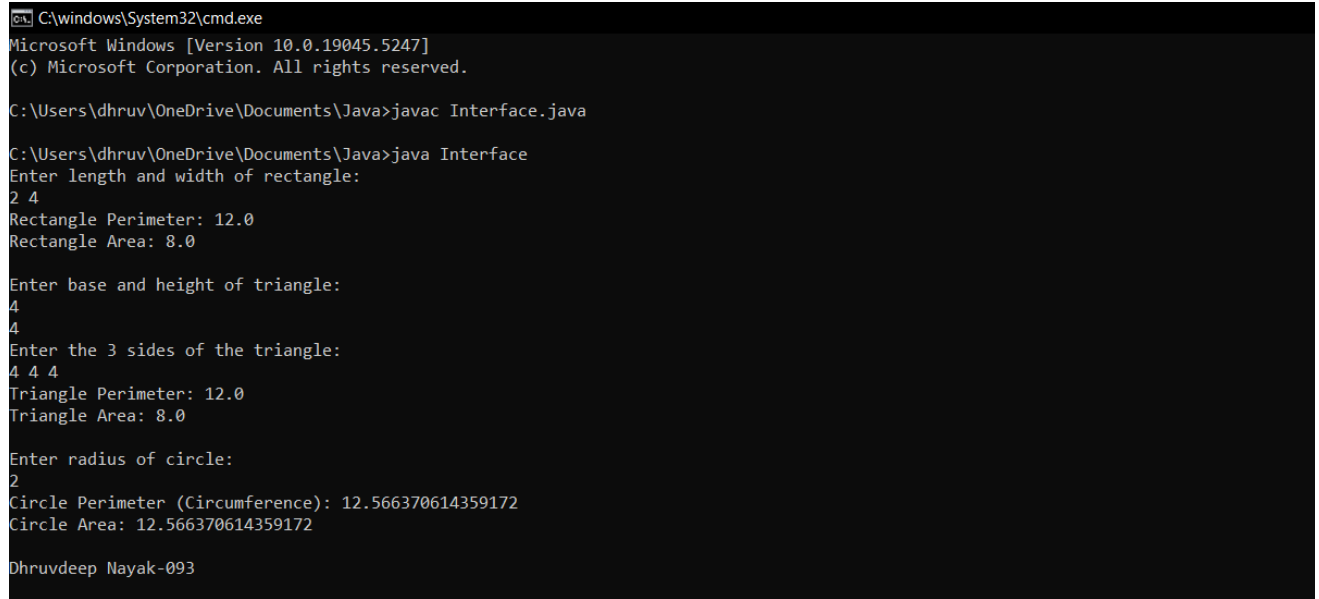
        System.out.println("Circle Perimeter (Circumference): " + circle.getPerimeter());
        System.out.println("Circle Area: " + circle.getArea());

        System.out.println("\nDhruvdeep Nayak-093");

        scanner.close();
    }
}

```

Output:



```

C:\windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac Interface.java

C:\Users\dhruv\OneDrive\Documents\Java>java Interface
Enter length and width of rectangle:
2 4
Rectangle Perimeter: 12.0
Rectangle Area: 8.0

Enter base and height of triangle:
4
4
Enter the 3 sides of the triangle:
4 4 4
Triangle Perimeter: 12.0
Triangle Area: 8.0

Enter radius of circle:
2
Circle Perimeter (Circumference): 12.566370614359172
Circle Area: 12.566370614359172

Dhruvdeep Nayak-093

```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program8>

Program 8:

Write a program that demonstrates handling of exceptions in inheritance tree. Create a base class called “Father” and derived class called “Son” which extends the base class. In Father class, implement a constructor which takes the age and throws the exception WrongAge() when the input age=father’s age.

Observation Book:

26/11/19

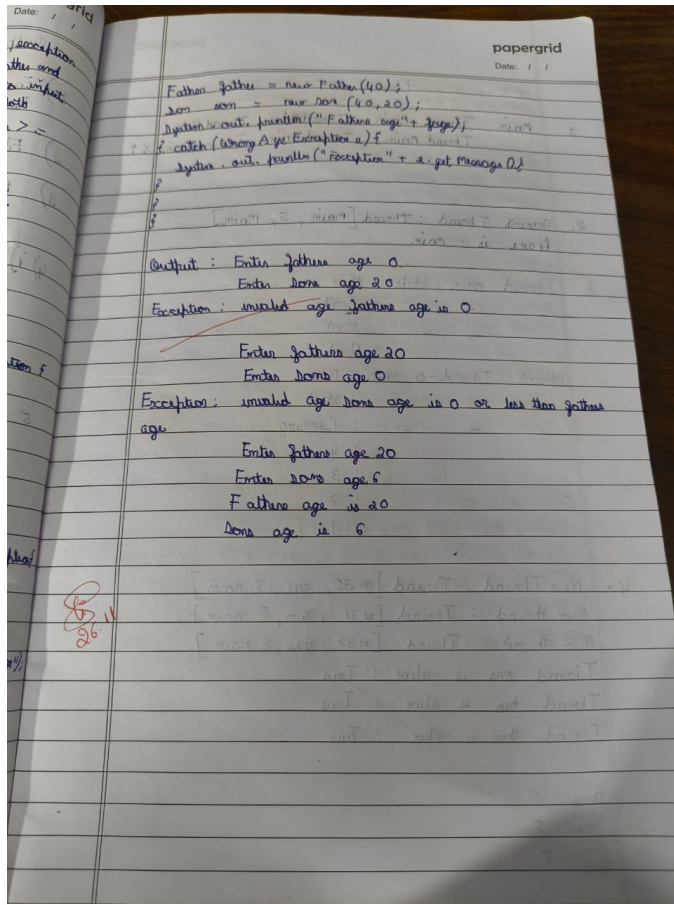
Write a program that demonstrates handling of exceptions in inheritance tree. Create a base class called Father and derived class Son. Father implements constructor that throws age < 0. Son does implement constructor that throws age > 100. Father and Son age throws an exception if some age > 100. Father age if some age is not greater than 100.

```
class WrongAgeException extends Exception {  
    public WrongAgeException(String message) {  
        super(message);  
    }  
}
```

```
class Father {  
    int age;  
    public Father(int age) throws WrongAgeException {  
        if (age < 0) throws new WrongAgeException("Father age cannot be 0");  
        this.age = age;  
    }  
}
```

```
class Son extends Father {  
    public Son(int fAge, int sAge) throws WrongAgeException {  
        super(fAge);  
        if (sAge < 0 || sAge > 100) {  
            throws new WrongAgeException("Invalid son age");  
        }  
    }  
}
```

```
public class Age {  
    public static void main (String[] args) {  
        try {  
            // ...  
        }  
    }  
}
```



Code:

```

class WrongAgeException extends Exception {
    public WrongAgeException(String message) {
        super(message);
    }
}

class Father {
    int age;

    public Father(int age) throws WrongAgeException {
        if (age < 0) {
            throw new WrongAgeException("Father's age cannot be negative");
        }
        this.age = age;
    }

    public void displayAge() {
        System.out.println("Father's age is: " + age);
    }
}

```

```

class Son extends Father {

    public Son(int fatherAge, int sonAge) throws WrongAgeException {
        super(fatherAge);
        if (sonAge >= fatherAge) {
            throw new WrongAgeException("Invalid son's age. Son's age should be less than father's
age.");
        }
        System.out.println("Son's age is: " + sonAge);
    }
}

public class Main {
    public static void main(String[] args) {
        try {
            Father father = new Father(40);
            Son son = new Son(40, 20);
            father.displayAge();
        } catch (WrongAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        try {
            Father father = new Father(20);
            Son son = new Son(20, 25); // This will throw an exception
        } catch (WrongAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        try {
            Father father = new Father(20);
            Son son = new Son(20, 6); // This is valid
            father.displayAge();
        } catch (WrongAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        System.out.println("Dhruvdeep Nayak-093");
    }
}

```

Output:

```
C:\Users\Dell\Desktop>javac JMain.java

C:\Users\Dell\Desktop>java JMain
Son's age is: 20
Father's age is: 40
Exception: Invalid son's age. Son's age should be less than father's age.
Son's age is: 6
Father's age is: 20
Dhruvdeep Nayak-093
```

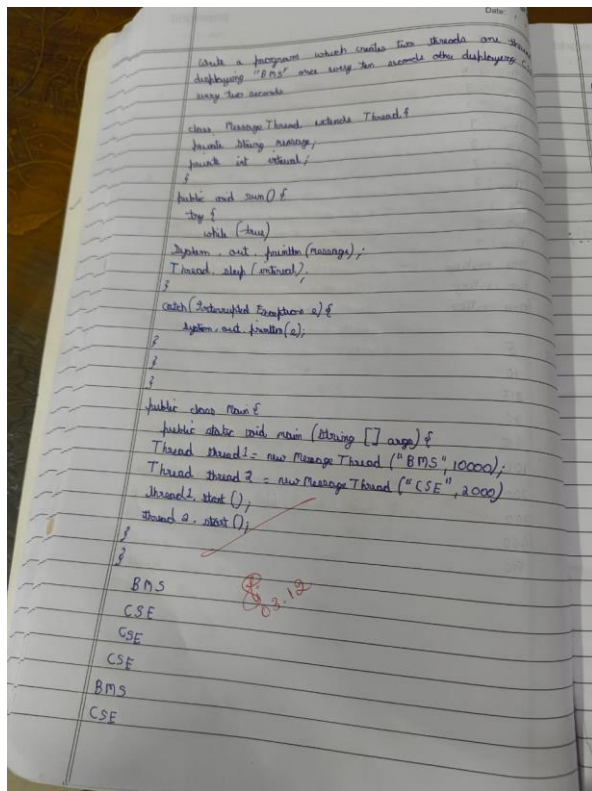
GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program9>

Program 9:

Write a program which creates two threads, one thread displaying “BMS” once every ten seconds and another displaying “CSE” once every two seconds.

Observation Book:



Code:

```
class MessageThread extends Thread {
    private String message;
    private int interval;

    public MessageThread(String message, int interval) {
        this.message = message;
        this.interval = interval;
    }

    @Override
    public void run() {
        try {
            while (true) {
                System.out.println(message);
                Thread.sleep(interval);
            }
        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }

    public static void main(String[] args) {
        Thread thread1 = new MessageThread("BMS", 10000);
        Thread thread2 = new MessageThread("SE", 2000);

        thread1.start();
        thread2.start();

        System.out.println("Dhruvdeep Nayak-093");
    }
}
```

Output:


```
C:\Windows\System32\cmd.exe - java MainThread
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhruv\OneDrive\Documents\Java>javac MainThread.java

C:\Users\dhruv\OneDrive\Documents\Java>java MainThread
Dhruvdeep Nayak-093
BMS
CSE
CSE
CSE
CSE
CSE
CSE
BMS
CSE
CSE
CSE
CSE
CSE
BMS
CSE
CSE
CSE
```

GitHub Link:

<https://github.com/DhruvdeepBMSCE/OOJ/tree/main/Program10>

Program 10:

Write a program that creates a user interface to perform integer divisions. The user enters two numbers in the text fields, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1 or Num2 were not an integer, the program would throw a NumberFormatException. If Num2 were Zero, the program would throw an Arithmetic Exception Display the exception in a message dialog box.

Observation Book:

one thread
laying out

papergrid
Date: / /

Basic calculator app

```

import javax.swing.*;
import java.awt.*;
import javax.swing.event.*;
import java.awt.event.*;

public class DivisionApp extends JFrame {
    JTextField num1Field, num2Field, resultField;
    JButton divideButton;

    public DivisionApp() {
        setTitle("Division App");
        setLayout(new FlowLayout());
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JLabel numLabel1 = new JLabel("Num 1");
        numField = new JTextField(10);

        JLabel numLabel2 = new JLabel("Num 2");
        numField = new JTextField(10);

        JLabel resultLabel = new JLabel("Result");
        resultField = new JTextField(10);
        resultField.setEditable(false);

        divideButton = new JButton("Divide");

        add(num1Label);
        add(num1Field);
        add(num2Label);
        add(num2Field);

```

```

add(divideButton);
add(resultLabel);
add(resultField);

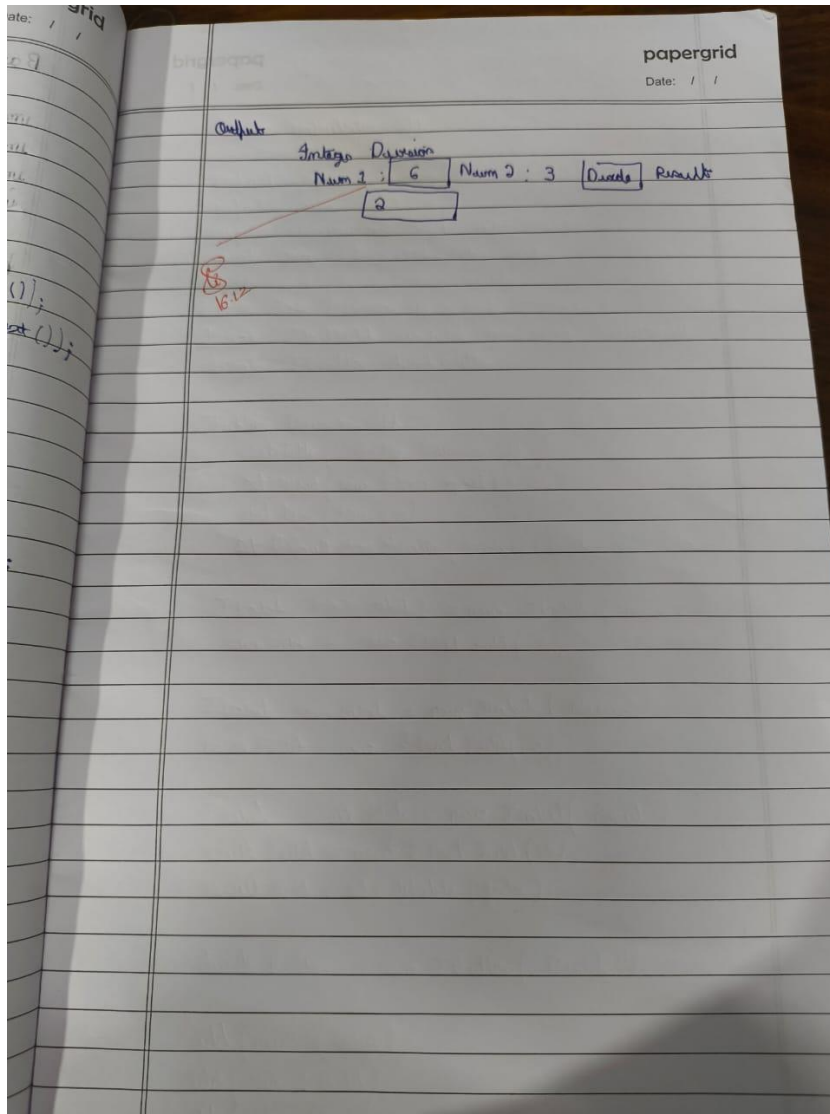
divideButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        try {
            num1 = Integer.parseInt(num1Field.getText());
            num2 = Integer.parseInt(num2Field.getText());
            int result = num1 / num2;

            resultField.setText(String.valueOf(result));
        } catch (NumberFormatException e) {
            JOptionPane.showMessageDialog(this, "Please Enter valid integer", "Error Message");
        }
    }
});

public static void main(String[] args) {
    SwingUtilities.invokeLater(new Runnable() {
        public void run() {
            new DivisionApp().setVisible(true);
        }
    });
}

```

(1) if test to user - hint dialog
 (2) if test to user - hint dialog
 (3) if test to user - hint dialog
 (4) if test to user - hint dialog



Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class DivisionApp extends JFrame {
    private JTextField num1Field, num2Field, resultField;
    private JButton divideButton;

    public DivisionApp() {
        setTitle("Integer Division App");
        setLayout(new FlowLayout());
        setSize(300, 200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```

```

JLabel num1Label = new JLabel("Num1:");
num1Field = new JTextField(10);

JLabel num2Label = new JLabel("Num2:");
num2Field = new JTextField(10);

JLabel resultLabel = new JLabel("Result:");
resultField = new JTextField(10);
resultField.setEditable(false);

divideButton = new JButton("Divide");

add(num1Label);
add(num1Field);
add(num2Label);
add(num2Field);
add(divideButton);
add(resultLabel);
add(resultField);

divideButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        try {
            int num1 = Integer.parseInt(num1Field.getText());
            int num2 = Integer.parseInt(num2Field.getText());

            int result = num1 / num2;
            resultField.setText(String.valueOf(result));

        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(DivisionApp.this, "Please enter valid integers.",
"Input Error", JOptionPane.ERROR_MESSAGE);
        } catch (ArithmeticException ex) {
            JOptionPane.showMessageDialog(DivisionApp.this, "Division by zero is not allowed.",
"Arithmetic Error", JOptionPane.ERROR_MESSAGE);
        } finally {

            System.out.println("Chethan K S\n1BM23CS074");
        }
    }
});
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(new Runnable() {

```

```
@Override
public void run() {
    new DivisionApp().setVisible(true);
}
});
}
```

Output:

